

What are the well-established methods for regularized covariance matrix estimation when the number of observations is smaller than the dimension (rank-deficient sample covariance),

with applications to recovering the underlying eigenvalue spectrum?

Regularized High-Dimensional Covariance & Spectrum Estimation: Main Method Families

When the number of variables exceeds or rivals the number of samples, the sample covariance is rank-deficient or ill-conditioned and its eigenvalues are heavily biased. Research has developed several regularization strategies that either impose structure on the covariance or directly correct the eigenvalue spectrum, often with the goal of recovering population eigenvalues.

Main Classes of Regularized Covariance Estimators

1. Structured (Model-Based) Covariance Estimation

- **Tapering/banding:** Enforce approximate banded structure (e.g., time/space ordering) and often combine with shrinkage toward identity, such as the **Tabasco** tapered–shrinkage estimator for low n/p ratios (Lam, 2019; Karoui, 2006).
- **Factor / spiked covariance models:** Use low-rank plus noise structure and adjust eigenvalues/eigenvectors; examples include S-POET for spiked models and factor-based methods (Lam, 2019; Wang & Fan, 2017).
- These methods improve conditioning and can target interpretable structure but rely on correct model assumptions (Lam, 2019; Karoui, 2006; Bartz, 2016).

2. Shrinkage Toward Simple Targets (Linear / Polynomial Shrinkage)

- **Linear shrinkage (Ledoit–Wolf and extensions):** Shrink sample covariance toward a multiple of the identity (or other targets) with analytically chosen weights to minimize mean squared error, effective in “large p , small n ” and under elliptical distributions (Lam, 2019; Won et al., 2013; Zhang et al., 2022; Ollila & Raninen, 2018; Chen et al., 2010).
- **Higher-order Stein-type shrinkage:** Model the estimator as a polynomial in the sample covariance and a target matrix, optimized via convex MSE criteria to yield well-conditioned estimators (Ledoit & Wolf, 2012).
- **Condition-number regularization:** Directly constrain the condition number using a likelihood-based approach, yielding a Steinian-type shrinkage estimator (Won et al., 2013).

3. Rotation-Invariant / Eigenvalue-Shrinkage Methods

Goal	Typical Approach	Population spectrum link	Citations
Better eigenspectrum	Nonlinear shrinkage of sample eigenvalues (keep eigenvectors)	Uses large-dimensional random matrix theory; approximates oracle eigenvalue map	(Lam, 2019; Won et al., 2013; Ledoit & Wolf, 2020; Jain et al., 2023; Ledoit & Wolf, 2020)

Goal	Typical Approach	Population spectrum link	Citations
Direct spectrum recovery	Solve a Marčenko–Pastur–based inverse problem for eigenvalue distribution	Estimates population eigenvalue distribution via convex optimization, then recovers eigenvalues	(Lam, 2019; Karoui, 2006; Jain et al., 2023; Banerjee & Monni, 2021)
Optimal shrinkers in spiked models	Design loss-specific optimal eigenvalue shrinkers	Closed forms under spiked covariance asymptotics	(Wang & Fan, 2017; Donoho et al., 2013)
Moment-based spectrum estimation	Recover moments then eigenvalues (even $n \ll p$)	Gives finite-sample error bounds, consistent even when $n/p \rightarrow 0$	(Kong & Valiant, 2016)

FIGURE 1 Key eigenvalue-based methods for spectrum estimation

Rotation-invariant approaches are central for **recovering the underlying eigenvalue spectrum**: they explicitly model the link between sample and population eigenvalues using random matrix theory (Marčenko–Pastur) and/or spiked models (Wang & Fan, 2017; Karoui, 2006; Ledoit & Wolf, 2020; Jain et al., 2023; Kong & Valiant, 2016; Banerjee & Monni, 2021; Donoho et al., 2013).

4. Bayesian and Robust Shrinkage

- **Bayesian gSIW prior**: Generalized shrinkage inverse-Wishart priors give posterior-consistent eigenvalues/eigenvectors under spiked covariance, even when $p/n \rightarrow \infty$ (Kim et al., 2025).
- **Robust shrinkage (Tyler-type)**: Regularized robust estimators handle heavy-tailed elliptical data and remain usable in $n < p$, with MMSE-based shrinkage selection (Chen et al., 2010).

Summary

Well-established tools include (i) structural regularizers (factor, tapering/banding), (ii) shrinkage toward simple targets (linear or higher-order), (iii) rotation-invariant eigenvalue shrinkage and spectrum inversion using Marčenko–Pastur and spiked models, and (iv) Bayesian and robust shrinkage priors. For **eigenvalue spectrum recovery**, nonlinear eigenvalue shrinkage and random-matrix-based spectrum estimators are the main workhorses, often providing consistent population eigenvalue estimates even when the sample covariance is rank-deficient.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Banerjee, S., & Monni, S. (2021). An Orthogonally Equivariant Estimator of the Covariance Matrix in High Dimensions and for Small Sample Sizes.. *Journal of statistical planning and inference*, 213, 16-32.
<https://doi.org/10.1016/j.jspi.2020.10.006>
- Bartz, D. (2016). Advances in high-dimensional covariance matrix estimation.
<https://doi.org/10.14279/depositonce-5038>

- Chen, Y., Wiesel, A., & Hero, A. (2010). Robust Shrinkage Estimation of High-Dimensional Covariance Matrices. *IEEE Transactions on Signal Processing*, 59, 4097-4107. <https://doi.org/10.1109/tsp.2011.2138698>
- Donoho, D., Gavish, M., & Johnstone, I. (2013). Optimal Shrinkage of Eigenvalues in the Spiked Covariance Model.. *Annals of statistics*, 46 4, 1742-1778. <https://doi.org/10.1214/17-aos1601>
- Jain, S., Krishnamurthy, V., Rangaswamy, M., Kang, B., & Gogineni, S. (2023). Radar Clutter Covariance Estimation: A Nonlinear Spectral Shrinkage Approach. *IEEE Transactions on Aerospace and Electronic Systems*, 59, 7640-7653. <https://doi.org/10.1109/taes.2023.3292069>
- Karoui, N. (2006). Spectrum estimation for large dimensional covariance matrices using random matrix theory. *Annals of Statistics*, 36, 2757-2790. <https://doi.org/10.1214/07-aos581>
- Kim, S., Lee, K., Park, S., & Lee, J. (2025). Eigenstructure inference for high-dimensional covariance with generalized shrinkage inverse-Wishart prior.
- Kong, W., & Valiant, G. (2016). Spectrum Estimation from Samples. *ArXiv*, abs/1602.00061. <https://doi.org/10.1214/16-aos1525>
- Lam, C. (2019). High-dimensional covariance matrix estimation. *Wiley Interdisciplinary Reviews: Computational Statistics*, 12. <https://doi.org/10.1002/wics.1485>
- Ledoit, O., & Wolf, M. (2012). Nonlinear Shrinkage Estimation of Large-Dimensional Covariance Matrices. *Annals of Statistics*, 40, 1024-1060. <https://doi.org/10.1214/12-aos989>
- Ledoit, O., & Wolf, M. (2020). Shrinkage estimation of large covariance matrices: Keep it simple, statistician?. *J. Multivar. Anal.*, 186, 104796. <https://doi.org/10.5167/uzh-172202>
- Ledoit, O., & Wolf, M. (2020). The Power of (Non-)Linear Shrinking: A Review and Guide to Covariance Matrix Estimation. *Journal of Financial Econometrics*. <https://doi.org/10.1093/jfinec/nbaa007>
- Ollila, E., & Raninen, E. (2018). Optimal Shrinkage Covariance Matrix Estimation Under Random Sampling From Elliptical Distributions. *IEEE Transactions on Signal Processing*, 67, 2707-2719. <https://doi.org/10.1109/tsp.2019.2908144>
- Wang, W., & Fan, J. (2017). Asymptotics of empirical eigenstructure for high dimensional spiked covariance.. *Annals of statistics*, 45 3, 1342-1374. <https://doi.org/10.1214/16-aos1487>
- Won, J., Lim, J., Kim, S., & Rajaratnam, B. (2013). Condition-number-regularized covariance estimation. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 75. <https://doi.org/10.1111/j.1467-9868.2012.01049.x>
- Zhang, B., Huang, H., & Chen, J. (2022). Estimation of Large-Dimensional Covariance Matrices via Second-Order Stein-Type Regularization. *Entropy*, 25. <https://doi.org/10.3390/e25010053>